Unified Harmonic Shells (UHS)

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Abstract

The Unified Harmonic Shells (UHS) framework establishes a radial quantization law derived by composing the curvature-frequency mapping of The Phi-Operator ($\Lambda^2 \cdot \Phi$) [2] with discrete harmonic boundary conditions, positioned at Tier Λ^3 (Global Properties) within the Stratified Axiomatics hierarchy [1]. The resulting sequence is calibrated against the atomic K-shell scale (the Bohr radius, a_0) and shown to admit a defined closure relation consistent with the Bohr model's $n^2 a_0$ radius scaling [3]. This paper establishes the formal radial quantization mechanism, the K-shell calibration and closure relation underlying it, and the scope boundaries required for downstream harmonic dependencies.

1. Foundational Premise

Shells are not passive spatial boundaries; they are candidate nodes of phase-locked harmonic resonance. In systems centered on a curvature-generating attractor, spatial distributions can exhibit discrete, concentric layering. UHS models this discretization as a geometric and energetic consequence of curvature-frequency duality. Radial shells form at spatial nodes where system curvature and temporal frequency constructively align to produce closed phase loops, establishing stable structural boundaries.

2. Core Mechanism: The Stratified Radial Law

UHS imports the differential mapping established by The Phi-Operator ($\Lambda^2 \cdot \Phi$) [2], which maps a characteristic curvature radius r to its temporal frequency $f(r)$:

$$f(r) = \Lambda^2 \cdot \Phi(r) = \frac{\Lambda^2}{r^{3/2}} \tag{1}$$

where Λ^2 is the system-specific dimensional scaling parameter $[L^{3/2}T^{-1}]$, classified under Tier Λ^2 (Relational Axioms) [1].

Quantization Derivation

Discrete shell layers emerge when the system's characteristic frequency locks into an allowed harmonic state f_n, governed by a fundamental mode f_0 and discrete harmonic order $n \in \mathbb{N}$ under the Tier Λ^1 constructive rule:

$$f_n = n \cdot f_0 \tag{2}$$

Setting $f(r_n) = f_n$ yields the quantized curvature-frequency relation:

$$\frac{\Lambda^2}{r_n^{3/2}} = n \cdot f_0 \tag{3}$$

Inverting for the quantized radial shell distance r_n:

$$r_n = \left(\frac{\Lambda^2}{n \cdot f_0}\right)^{2/3} \tag{4}$$

Under this relation, r_n decreases as n increases: higher harmonic order corresponds to tighter binding at smaller radius, consistent with denser curvature at higher-frequency states.

Baseline Anchor (r_0)

Structure begins at $n = 1$. The quantity $r_0 = c/f_0$ is not the $n = 0$ member of the shell sequence r_n; it is the nonzero baseline scale from which harmonic structure ($n = 1, 2, 3, \ldots$) is generated, where c denotes the system phase velocity constant $[L \cdot T^{-1}]$ inherited from the base manifold. This baseline allows downstream frameworks built on non-zero starting references to close without encountering zero-division. r_0 and r_1 are related through f_0 but are not the same quantity, and this paper does not claim they are.

2.1. K-Shell Calibration and Defined Boundary Closure

The atomic K-shell provides the reference calibration for the radial quantization sequence. The resulting sequence is calibrated by setting the $n = 1$ internal node to the Bohr radius a_0, fixing f_0 for the atomic case:

$$r_n = \left(\frac{\Lambda^2}{n \cdot f_0}\right)^{2/3} = \frac{a_0}{n^{2/3}} \tag{5}$$

which specifies the internal 1D radial curvature-node density within the central potential well: K-shell ($n = 1$) at $1.000\,a_0$, L-shell ($n = 2$) at $0.630\,a_0$, M-shell ($n = 3$) at $0.480\,a_0$, N-shell ($n = 4$) at $0.397\,a_0$. Here r_n represents the internal 1D standing-wave curvature node density within the central potential, distinct from the projected 3D spatial shell boundary introduced next.

Within the UHS framework, the outer spatial boundary $\langle r_n \rangle$ is defined in terms of the internal curvature node r_n via the structural closure relation:

$$\langle r_n \rangle = a_0 \cdot \left(\frac{a_0}{r_n}\right)^3 = a_0 \cdot n^2 \tag{6}$$

This closure is presented as a defined structural identity within UHS — not as an independent first-principles derivation of the Bohr model's $n^2 a_0$ radius scaling [3], which is established separately. The claim is precisely this: *given* the UHS internal radial law, this closure transformation maps it onto the Bohr model's established $n^2 a_0$ scaling. UHS does not derive the K-shell; it calibrates against it and shows the resulting sequence closes consistently.

In this atomic calibration, the formal harmonic shell is defined at the outer phase-closed spatial boundary $\langle r_n \rangle$, anchored internally by the radial curvature node r_n. Beyond this calibrated case, the general radial law of Section 2 produces r_n alone; the $\langle r_n \rangle$ closure is specific to systems calibrated against a known outer-boundary scale such as a_0.

Shell	n	f_n	r_n (internal node)	$\langle r_n \rangle$ (Bohr-model scaling)
K	1	f_0	$1.000\,a_0$	$1\,a_0$
L	2	$2f_0$	$0.630\,a_0$	$4\,a_0$
M	3	$3f_0$	$0.480\,a_0$	$9\,a_0$
N	4	$4f_0$	$0.397\,a_0$	$16\,a_0$

3. Shell Conditions and Stability

Within UHS, a radial layer is classified as a candidate stable harmonic shell when it satisfies three tier-stratified constraints:

Frequency Quantization (Λ^1): $f_n \in \{n \cdot f_0 \mid n \in \mathbb{N}\}$

Curvature Matching (Λ^2): $\chi_n = 1/r_n = (n \cdot f_0/\Lambda^2)^{2/3}$, where χ_n denotes the UHS radial curvature measure, defined as $\chi_n \equiv 1/r_n$.

Phase Closure (Λ^3): $\oint_C k(r)\,dr = 2\pi m$, evaluated along a closed phase path C encircling the central attractor (or radial standing-wave boundary under WKB action quantization), where $k(r)$ is the local wavenumber and $m \in \mathbb{Z}$.

These three conditions are the classification criteria of the framework: a radial layer failing one or more does not satisfy the UHS shell condition at that radius.

4. Predictive Alignment and Drift Prevention

UHS specifies a deterministic classification rule based on tier-locked phase closure, in place of fitting shell structure to a continuous function after the fact:

Structural Boundary Classification: UHS identifies candidate stable spatial interfaces at phase-locked curvature-frequency nodes ($f_n = n \cdot f_0$).

Drift Prevention: Semantic and numerical drift can arise when continuous spatial functions are discretized without preserving the nonzero baseline r_0 or the discrete tier boundaries (Λ^0–Λ^3). UHS therefore preserves these quantities explicitly when constructing shell sequences. (For instance, fitting the atomic shell sequence to a continuous power law without preserving the discrete $n^2 a_0$ scaling specified by the Bohr model [3] would predict shell radii that diverge from the calibrated K, L, M, N sequence in Section 2.1.)

Non-Resonant Gaps: Spatial regions that do not satisfy the phase-closure condition are classified by UHS as non-resonant regions rather than harmonic shell locations.

5. Integration with Stratified Axiomatics

UHS operates as a higher-tier framework within the formal foundational hierarchy [1]:

Λ^0 **(Foundational Base):** Defines the underlying non-flat geometric base space and scalar fields.

Λ^1 **(Constructive Rules):** Generates discrete harmonic modes $f_n = n \cdot f_0$.

Λ^2 **(Relational Operators):** Applies The Phi-Operator [2] to map spatial curvature to frequency.

Λ^3 **(Global System Shells):** Enforces global phase closure and computes the nested field architecture r_n, calibrated in Section 2.1.

This tier hierarchy is intended to prevent category errors and semantic drift during application, consistent with the drift-prevention criteria established in Stratified Axiomatics [1].

Conclusion

Unified Harmonic Shells establishes a scale-invariant radial quantization law by composing the harmonic quantization rule of Λ^1 with the curvature-frequency mapping of The Phi-Operator ($\Lambda^2 \cdot \Phi$) [2], within the Stratified Axiomatics hierarchy [1], calibrated against the atomic K-shell scale and shown to admit a defined closure consistent with the Bohr model [3]. This paper establishes that mathematical foundation only; application beyond the atomic calibration case is left to dependent papers.

References

[1] Johnson, C. (2025). *Stratified Axiomatics: A Formal Mathematical Framework*. Series: Mathematical Foundations for Universal Systems. SemanticDrift.

[2] Johnson, C. (2025). *The Phi-Operator ($\Lambda^2 \cdot \Phi$): A Differential Map from Curvature to Frequency*. Series: Mathematical Foundations for Universal Systems. SemanticDrift.

[3] Bohr, N. (1913). *On the Constitution of Atoms and Molecules*. Philosophical Magazine, 26(1), 1–25.